RESTAURANT RECOMMENDATION SYSTEM

1. PROJECT OVERVIEW

The Restaurant Recommendation System is a machine learning-based web application developed to assist users in discovering restaurants based on their preferences, tastes, and dining patterns. With the rapid growth of food delivery platforms and dining options, users often face difficulty in selecting suitable restaurants. This system addresses that challenge by providing intelligent recommendations.

The project uses Content-Based Filtering, where recommendations are generated by analyzing restaurant attributes such as cuisines, ratings, and cost. The system identifies similarities between restaurants and suggests alternatives that closely match the user's choice.

The application is implemented using Flask for backend processing and a web-based interface for user interaction. Users input a restaurant name, and the system returns a list of similar restaurants ranked based on similarity scores.

This project demonstrates the integration of machine learning with web development, providing a practical and scalable solution for real-world recommendation problems.

Project Member:

Atharva Pramodkumar Kumbhar

2. OBJECTIVES

- To design and develop a recommendation system using content-based filtering techniques
- To preprocess and clean real-world restaurant datasets for analysis
- To extract meaningful features such as cuisines, ratings, and cost for modeling
- To implement TF-IDF vectorization for text-based similarity computation
- To compute similarity between restaurants using cosine similarity
- To build a responsive and user-friendly web application using Flask
- To provide personalized restaurant recommendations based on user input
- To evaluate system performance and improve recommendation accuracy
- To demonstrate the application of machine learning in real-world scenarios

3. KEY FEATURES

- 🔍 Search-based recommendation system
- 🍴 Suggests similar restaurants based on cuisines
- ⭐ Displays ratings and cost for better decision-making
- ⚡ Fast and efficient recommendation engine
- 🌐 Web-based interface using Flask
- 📊 Data preprocessing and feature extraction
- 📐 Uses TF-IDF and cosine similarity

4. USE CASE SCENARIOS

Scenario 1: Restaurant Visitors

Users looking for dining options can input their favorite restaurant and receive recommendations for similar restaurants. This improves user satisfaction by helping them discover new places aligned with their taste preferences.

Scenario 2: Restaurant Owners

Restaurant businesses can benefit from recommendation systems by being suggested to users who prefer similar cuisines. This increases visibility and customer reach.

Scenario 3: Food Delivery Platforms

Online food delivery services can integrate such systems to recommend restaurants based on user preferences, leading to increased engagement and higher order volumes.

Scenario 4: Tourists & Travelers

Tourists visiting a new city can use the system to explore restaurants similar to their known preferences, making it easier to find suitable dining options.

5. TECHNICAL APPROACH

Frontend (User Interface)

- Developed using HTML, CSS, and JavaScript
- Provides input forms for restaurant search
- Displays recommendations in structured format (tables/cards)
- Designed for simplicity and ease of use

Backend (Flask Framework)

- Handles user requests and processes input data
- Communicates with machine learning model
- Returns processed results to frontend
- Uses routing and templates for page rendering

Machine Learning Model

- Technique: Content-Based Filtering
- Feature Extraction: TF-IDF Vectorization
- Output: Similar restaurant list

Data Processing

- Removal of null values and duplicates
- Selection of relevant features (name, cuisines, rating, cost)

System Architecture

- Presentation Layer: User interface (HTML/CSS)
- Application Layer: Flask backend logic

6. IMPLEMENTATION PLAN

Phase	Activities	Timeline
Requirement Analysis	Define objectives, dataset, and use cases	2 Days
Design & Architecture	Design system workflow and UI layout	3 Days
Data Preprocessing	Clean dataset and prepare features	3 Days
Model Development	Implement TF-IDF and similarity model	3 Days
Frontend Development	Build UI using HTML/CSS	5 Days
Backend Development	Integrate Flask with ML model	4 Days
Testing & Debugging	Fix errors and validate output	3 Days
Deployment	Run application locally or on cloud	2 Days

7. BENEFITS

FOR USERS

- Saves time in searching for restaurants
- Provides personalized and relevant suggestions
- Improves overall dining experience

FOR BUSINESSES

- Increases restaurant visibility
- Helps target the right audience
- Enhances customer engagement

FOR PLATFORMS

- Improves user retention
- Increases order conversion rates
- Provides intelligent recommendation systems

8. PROJECT FLOW

- User enters a restaurant name through the web interface
- The input is sent to the Flask backend
- The system processes the dataset and extracts features
- TF-IDF vectorization converts text data into numerical form
- Cosine similarity calculates similarity between restaurants
- Top similar restaurants are selected
- Results are sent back to frontend
- Recommendations are displayed to the user

9. REQUIREMENTS SPECIFICATION

System Requirements

- Python 3.8 or higher
- Operating System: Windows / Linux / macOS
- Minimum RAM: 4GB (recommended 8GB)
- Web browser (Chrome, Edge, etc.)

Software & Libraries

- pandas (data processing)
- numpy (numerical operations)
- scikit-learn (ML algorithms)
- flask (web framework)

INSTALLATION STEPS :

pip install pandas numpy scikit-learn flask

Run Application

python app.py

10. RISKS AND MITIGATIONS

Risk	Impact	Mitigation Strategy
Large dataset size	High memory usage	Use optimized similarity computation
Data inconsistency	Incorrect recommendations	Clean and preprocess dataset
Performance issues	Slow response time	Optimize algorithms
Limited features	Reduced accuracy	Add more features (location, reviews)
User input errors	No results found	Implement validation

11. FUTURE ENHANCEMENTS

- Location-based recommendations
- Hybrid recommendation system (content + collaborative filtering)
- Mobile application integration
- User reviews and ratings system
- Advanced filtering (price, rating, cuisine)
- Multi-city dataset expansion
- Deep learning-based recommendation models

12. CONCLUSION

The Restaurant Recommendation System successfully demonstrates how machine learning can be applied to solve real-world problems. By leveraging content-based filtering techniques, the system provides personalized restaurant suggestions efficiently.

The integration of Flask and machine learning showcases a complete end-to-end application, from data processing to user interaction. This project highlights the importance of intelligent systems in enhancing user experience and decision-making.

With further improvements, this system can be expanded into a full-scale recommendation platform used by businesses and consumers alike.